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Roll No

CE-402 (GS)
B.E. IV Semester Examination, June 2020
Grading System (GS)
Concrete Technology
Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.
ii) All questions carry equal marks.

1. Explain the procedure for determination of 'ten percent fines value'.
2. Discuss ACI mix design method.
3. What are the air-entraining admixtures? How do they affect water cement ratio, segregation and bleeding and permeability of concrete?
4. Explain the term "Grade" of concrete. Write the various advantages and disadvantages of concrete.
5. Explain micro-cracking of concrete in detail.
6. Write a short note on the feasibility of use of sea water for mixing concrete.

OR

What is meant by ordinary Portland cement 53 grade? What are the chemical composition of Portland cement?

7. What is Ferrocement concrete? What are its uses, advantages and disadvantages?

OR

Write a brief procedure on how to determine the initial setting time of cement.

8. Answer any four of the following:
 - a) What factors control the proportion of concrete mix?
 - b) Explain stability, mobility and compatibility.
 - c) Give acceptance criteria for concrete.
 - d) Discuss different ways of compaction of concrete.
 - e) Discuss uses of SIFCON and its effects.
